Internet & Prior-Art Research Report

Multi-Model Evidence-Grounded Answering for a Sovereign Agentic AI Workbench | Prepared 30 October 2026

1. Research question

Has anyone built the same thing: multiple local language models answering one question over a shared, workspace-scoped document retrieval, with measurable agreement, source-constrained merging, and enterprise audit controls? This report summarizes what publicly exists, what each prior work does and does not do, and where the gap is.

2. Short answer

The individual ingredients all exist in the literature; the combination does not. No system found in the literature combines: (a) one shared workspace-scoped retrieval, (b) multiple different local models answering independently, (c) deterministic (non-trained, non-ML) agreement scoring exposed to a human, (d) synthesis re-grounded in the sources with enforced citations, (e) claim-level verification, inside (f) an audited, RBAC-controlled, on-premise-only control plane.

3. Prior work: what exists on the internet

Eight closely related works were identified and analyzed. For each: what they found or discovered, how their system works, what is better in their approach, and how this project tackles the difference.

Mixture-of-Agents (MoA)

Reference: Wang et al., arXiv:2406.04692 (2024); Together AI implementation
What they found / discovered / updated: DISCOVERED: an ensemble of open-source LLMs arranged in layers can beat a far stronger monolithic model - 65.1% on AlpacaEval 2.0 vs GPT-4 Omni's 57.5%. Follow-up work confirmed and extended the idea: Distributed MoA (edge devices gossiping answers, with proven queue stability), SMoA (sparse variant cutting cost), and Gated MoA (filtering semantically inconsistent intermediate outputs raises quality).
How their system works: MECHANISM: layered architecture. Layer 1 holds several proposer LLMs that each answer the prompt independently. Layer 2 (and any further layers) are aggregator LLMs: each one reads ALL previous-layer outputs as auxiliary information and produces a refined response. The final layer output is the answer. Diversity of proposers + aggregation is the engine of the gain.
What is better in theirs: WHAT IS BETTER IN THEIRS: benchmark-proven large gains; layered refinement lets models critique and build on each other, which one-shot fan-out cannot do; mature reference implementation and broad replication; works for open-ended generation where no ground sources exist.
How we tackle it: HOW WE TACKLE IT: (1) Latency - layers are sequential, so we flatten to ONE parallel fan-out round sized for a single local endpoint; (2) Grounding - MoA has no retrieval, so every model in our pipeline answers from the SAME numbered evidence block and synthesis is constrained to the sources; (3) Trust - we add a deterministic agreement matrix and claim-level verification, which MoA lacks entirely; (4) Sovereignty - MoA assumes cloud-scale serving; ours runs on one Ollama/vLLM/LM Studio endpoint with cloud calls refused.

LLM-Blender

Reference: Jiang, Ren & Lin, ACL 2023 (DOI 10.18653/v1/2023.acl-long.792)
What they found / discovered / updated: DISCOVERED: the best LLM differs per input - on their MixInstruct benchmark no single model wins consistently (e.g. Vicuna tops 21.2% of examples, ChatGLM 8.5%, FLAN-T5 under 1%), so ensembling beats any fixed choice. Also showed a trained pairwise ranker correlates best with ChatGPT-based ranking of outputs.
How their system works: MECHANISM: two modules. PairRanker jointly encodes the input and a PAIR of candidate outputs with cross-attention and scores which is better; it produces a ranking over all candidates. GenFuser then generates a fused output from the top-ranked candidates, capitalizing on their strengths and mitigating weaknesses.
What is better in theirs: WHAT IS BETTER IN THEIRS: a learned ranker is far more discriminative than lexical overlap - it understands semantic superiority, not just word overlap; the framework is general and model-agnostic; strong measured gains over every individual LLM and baseline.
How we tackle it: HOW WE TACKLE IT: (1) No training data or GPU budget on-premise - our Jaccard agreement matrix is the training-free substitute, and we accept it is a triage signal, not a judge; (2) Their fusion is unconstrained, ours is source-constrained: the synthesizer must drop anything no source supports and cite [n]; (3) Their ranker hides its reasoning - our pairwise overlaps are shown to the human reviewer as numbers.

RAG Ensemble

Reference: Chen, Dong, Zhu & Dou, arXiv:2508.13828 (2025)
What they found / discovered / updated: DISCOVERED: the first information-entropy explanation of WHY aggregating multiple RAG systems helps, plus systematic evidence that aggregation is generalizable and robust at BOTH the pipeline level (Branching, Iterative, Loop, Agentic) and the module level (Generator, Retriever, Reranker), across seven research questions.
How their system works: MECHANISM: run several RAG variants on the same query and aggregate their outputs or module choices - e.g. branch to multiple pipelines and merge answers, or ensemble retrievers and rerank the union. The entropy analysis explains that diverse systems reduce the uncertainty of the combined output.
What is better in theirs: WHAT IS BETTER IN THEIRS: theoretical grounding (entropy) rather than heuristics; breadth - they study four pipeline types and three module types systematically; evidence that module-level ensembling (e.g. of retrievers) also pays off.
How we tackle it: HOW WE TACKLE IT: (1) Scope - they ensemble whole RAG pipelines; we hold retrieval fixed and ensemble the GENERATOR dimension only, which keeps one auditable evidence trail instead of N divergent ones; (2) Deployment - their work is a research prototype; ours embeds the ensemble in a production control plane with RBAC, approval gates and append-only audit; (3) We adopt their core finding (aggregation helps) as validation of our design choice.

LLM Ensemble survey

Reference: Chen et al., IJCAI 2026 (arXiv:2502.18036)
What they found / discovered / updated: DISCOVERED/CONSOLIDATED: the field's first systematic taxonomy - ensemble-BEFORE-inference (route or select), ensemble-DURING-inference (query multiple models), ensemble-AFTER-inference (aggregate outputs) - plus a benchmark and application survey, and open research directions.
How their system works: MECHANISM: survey contribution - it organizes existing methods by WHEN the ensemble decision happens and by aggregation unit (token, sentence, passage, answer), and maps methods to benchmarks.
What is better in theirs: WHAT IS BETTER IN THEIRS: completeness and authority - it is the reference map of the field; it catalogs cost-aware and routing methods (e.g. picking the cheapest sufficient model) that our fixed fan-out does not attempt.
How we tackle it: HOW WE TACKLE IT: (1) Positioning - our method is ensemble-during-inference with an after-inference synthesis step, and we say so in exactly their taxonomy terms; (2) Their open problem 'aggregation needs a judge or trained fuser' is answered in our design by a deterministic agreement matrix plus a source-constrained synthesizer; (3) Cost-aware routing is future work we now have a map for.

Self-RAG family

Reference: Asai et al. 2023; successors incl. Reflective RAG (Wu et al., Findings of ACL 2026)
What they found / discovered / updated: DISCOVERED: a model can be trained to critique ITSELF - judging whether retrieval was needed, whether retrieved passages are relevant, and whether its own output is supported - and that these reflection signals improve factuality. Reflective RAG added RL-optimized strategy selection with reflection tagging, outperforming agentic baselines on five QA benchmarks.
How their system works: MECHANISM: the generator emits special reflection tokens (or a trained critique head) alongside content: ISREL (is the passage relevant), ISSUP (is the output supported), ISUSE (is it useful). Decoding is guided by these self-judgments; Reflective RAG trains the tagging with SFT then optimizes strategy with RL and KL regularization.
What is better in theirs: WHAT IS BETTER IN THEIRS: fine-grained, per-claim introspection during decoding; no extra models at inference time (the judgments come from the same model); strong published benchmark results.
How we tackle it: HOW WE TACKLE IT: (1) Training - Self-RAG requires training the model to emit reflection tokens, impossible with off-the-shelf Ollama models; our cross-model consensus needs zero training; (2) Self-grading weakness - a model judging its own output shares its own blind spots; a DIFFERENT model family disagreeing is a stronger error signal, and our verifier is a separate model checking claims against sources; (3) We keep their key insight - judge the output against the evidence - but move the judging outside the generator.

Self-Consistency

Reference: Wang et al., 2022 (arXiv:2203.11171)
What they found / discovered / updated: DISCOVERED: sampling the SAME model many times at high temperature and majority-voting the final answers dramatically improves accuracy on arithmetic and commonsense reasoning - diversity of reasoning paths, not model diversity, is enough.
How their system works: MECHANISM: sample N chain-of-thought completions from one model with temperature; extract the final answer from each; take the majority answer. Requires answers that reduce to a canonical form (a number, a choice) for voting.
What is better in theirs: WHAT IS BETTER IN THEIRS: extreme simplicity - one model, no extra infrastructure; large gains on reasoning benchmarks; well understood statistically.
How we tackle it: HOW WE TACKLE IT: (1) Answer format - majority voting fails on free-text cited answers, which is what our workbench produces; we merge GENERATIVELY with a synthesizer instead of voting; (2) Diversity source - sampling noise of one model misses systematic blind spots; we use genuinely different model families at low temperature, so errors are less correlated; (3) Cost shape - they spend N samples of one small model; we spend N models once, which suits serving several models on one endpoint.

Iterative Consensus Ensemble (ICE)

Reference: 2025, Computers in Biology and Medicine (S0010482525010820)
What they found / discovered / updated: DISCOVERED: letting multiple LLMs critique each other over ITERATIVE rounds and converge toward consensus refines answers across four datasets - models scrutinizing each other's outputs beats one-shot aggregation in their setting.
How their system works: MECHANISM: round 1: models answer independently. Round 2..k: each model sees the others' answers and critiques/revises its own. Repeat until answers stabilize or a round budget is spent; the consensus output is returned.
What is better in theirs: WHAT IS BETTER IN THEIRS: consensus quality - revision rounds let errors be caught and fixed before output; the final answer carries the weight of explicit cross-examination.
How we tackle it: HOW WE TACKLE IT: (1) Latency - each round multiplies generation time, unacceptable on one local endpoint; we do ONE round; (2) Evidence drift - critique text is itself ungrounded generation and can pull answers away from the sources; we give models NO channel to each other - only the synthesizer sees all answers, and it is re-grounded in the evidence; (3) Audit - N rounds of mutating answers are hard to audit; our single round with recorded per-model answers is fully auditable.

Multi-Agent RAG (MA-RAG)

Reference: Springer, DOI 10.1007/978-3-032-05179-0_22
What they found / discovered / updated: DISCOVERED: decomposing the RAG process into SPECIALIZED agents - query reformulation, iterative retrieval refinement, hallucination detection, answer validation - improves answer quality over monolithic RAG.
How their system works: MECHANISM: a pipeline of role-specialized agents: one rewrites the query, one iteratively refines retrieval, one detects hallucinations in drafts, one validates the final answer. Each agent can be prompted or fine-tuned for its role.
What is better in theirs: WHAT IS BETTER IN THEIRS: role specialization - a dedicated hallucination detector can be much better at that one job than a generalist; the architecture is modular and each stage is individually improvable.
How we tackle it: HOW WE TACKLE IT: (1) Different axis - they specialize ROLES, we replicate GENERALISTS: same evidence, N independent full answerers, consensus measured; (2) We still adopt their validation idea - our grounding verifier plays the 'answer validation' agent role, checking the merged answer claim-by-claim; (3) Operational simplicity - N specialist prompts and handoffs are complex to run and audit; our fan-out is one uniform call pattern repeated.

4. What the project's own documents said before this feature

- Research_Paper.pdf (manuscript draft, 8 Sep 2026): the documented system is SINGLE-model - evidence-gated local retrieval (thresholds 0.30 / 0.45), one generation, citations, owner approval for the report tool, append-only audit.
- The paper's Discussion (p. 9) explicitly flags the gap this feature fills: 'Model routing or larger models should be added only with measurements showing their benefit for that workload.'
- The paper's API load results (20,000 requests per stage, zero errors, p95 7.5-102 ms across 1-100 concurrent clients) cover the API layer, not generation; multi-model fan-out multiplies generation load and needs its own measurement.
- SIH_Documentation.pdf (dossier v1.0, 32 pages) documents the same single-model architecture in deployment terms.

5. Source list

- MoA: Wang et al., 'Mixture-of-Agents Enhances Large Language Model Capabilities', arXiv:2406.04692 (2024). https://arxiv.org/abs/2406.04692 - code: https://github.com/togethercomputer/moa
- LLM-Blender: Jiang, Ren & Lin, ACL 2023, DOI 10.18653/v1/2023.acl-long.792. https://github.com/avnlp/llm-blender
- RAG Ensemble: Chen, Dong, Zhu & Dou, 'Revisiting RAG Ensemble', arXiv:2508.13828 (2025). https://arxiv.org/abs/2508.13828
- LLM Ensemble survey: Chen et al., 'Harnessing Multiple Large Language Models: A Survey on LLM Ensemble', IJCAI 2026, DOI 10.24963/ijcai.2026/867. https://arxiv.org/abs/2502.18036
- Ensemble LLMs survey: Mienye & Swart, Information 16(8):688 (2025), DOI 10.3390/info16080688
- Reflective RAG: Wu et al., Findings of ACL 2026, DOI 10.18653/v1/2026.findings-acl.648
- ICE: 'Refining LLMs outputs with iterative consensus ensemble', Computers in Biology and Medicine (2025). https://www.sciencedirect.com/science/article/pii/S0010482525010820
- MA-RAG: Springer, DOI 10.1007/978-3-032-05179-0_22
- Self-Consistency: Wang et al., arXiv:2203.11171 (2022) (standard reference).
- RAG survey: Zhao et al., Data Science and Engineering (2026), DOI 10.1007/s41019-025-00335-5
- Project documents: Research_Paper.pdf and SIH_Documentation.pdf (8 Sep 2026 drafts, repository root).